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A shifted interface approach for internal discontinuities in poroelastic media

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Why Poroelastic Interfaces?

Where they arise

- Cartilage, bone, hydrogels
- Fractured geomaterials
- Subsurface faults and reservoirs

Why they are difficult

- Thin, complex, evolving geometry
- Strong flow–deformation coupling

Why they matter

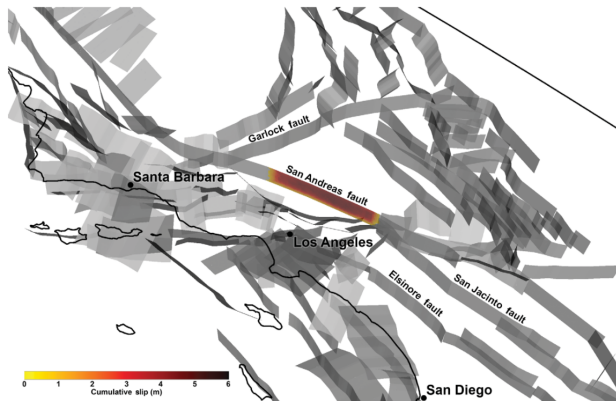
- CO₂ storage, geothermal, induced seismicity

We need methods that handle complex interfaces without remeshing.



Major faults and historic ruptures along the
San Andreas Fault System (USGS)

Fault Complexity



Source: Milner et al. (2021).

Real interfaces are irregular, intersecting, and geometrically complex

Where Standard Methods Pay the Price

Each route avoids one difficulty, but pays a different implementation burden.

Body-fitted FEM

Geometry follows the crack

Main burden

Remeshing around complex or evolving interfaces

XFEM

Fixed background mesh

Main burden

Enrichment management and special quadrature

CutFEM

Unfitted standard FE spaces

Main burden

Cut-cell integration and stabilization tuning

SIM keeps the mesh handling simple and relocates the complexity into transfer operators.

Our Approach: Shifted Interface Method

Goal: avoid remeshing, enrichment, and cut-cell complexity.

What SIM does

- Uses a mesh-aligned surrogate crack
- Transfers interface laws from surrogate to the true crack

Why it helps

- No remeshing around complex interfaces
- Standard FE data structures

Shift the laws, not the mesh.

Governing Equations

Unknowns

- p : pore pressure
- u_i : displacement

Quasi-static Biot
poroelasticity
in $\Omega \setminus \Gamma_c$

Flow balance

$$\beta \dot{p} + \alpha \dot{u}_{i,i} = -q_{i,i} + f$$

Momentum balance

$$\sigma_{ij,j} = 0$$

$$\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2G \varepsilon_{ij} - \alpha p \delta_{ij}, \quad q_i = -\gamma p_{,i}, \quad \gamma = k/\eta$$

Standard poroelasticity in the bulk — difficulty lies at the interface.

Interface Conditions on the True Crack Γ_c

Jump & average operators:

$$[\![\phi]\!] := \phi^+ - \phi^-, \quad \{\{\phi\}\} := \frac{1}{2}(\phi^+ + \phi^-)$$

Reference normal on Γ_c :

$$\hat{n}_i^+ = +\hat{n}_i, \quad \hat{n}_i^- = -\hat{n}_i$$

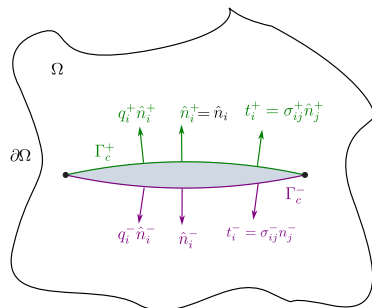
Momentum balance and spring law:

$$[\![\hat{\sigma}_{ij}]\!] \hat{n}_j = 0, \quad \{\{\hat{\sigma}_{ij}\}\} \hat{n}_j = K_{ij} [\![\hat{u}_j]\!] + H_{ij} [\![\hat{\dot{u}}_j]\!]$$

Mass balance and Robin law:

$$[\![\hat{q}_i]\!] \hat{n}_i = 0, \quad \{\{\hat{q}_i\}\} \hat{n}_i = -T_n [\![\hat{p}]\!]$$

One-sided interface unknowns are introduced later only for enforcement.



Two-sided traces on Γ_c with a common reference normal $\hat{n}_i$

Broken Variational Form: Where the Difficulty Appears

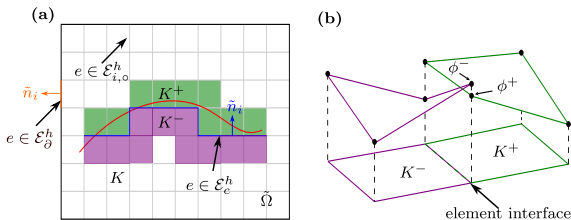
After elementwise integration by parts on $\tilde{\Omega}$:

Only the boxed face terms need crack-specific modeling.

$$\begin{aligned} 0 = & \int_{\tilde{\Omega}} (\beta \dot{p} + \alpha \dot{u}_{k,k}) v \, d\tilde{\Omega} + \int_{\tilde{\Omega}} \gamma p_{,i} v_{,i} \, d\tilde{\Omega} \\ & + \int_{\tilde{\Omega}} \sigma_{ij} w_{i,j} \, d\tilde{\Omega} + \text{standard outer-boundary and source terms} \\ & - \boxed{\mathcal{I}_{\mathcal{E}_i^h}^{\text{flow}}(p, v)} + \boxed{\mathcal{I}_{\mathcal{E}_i^h}^{\text{mech}}(u, p, w)} \end{aligned}$$

Everything else is standard FE machinery away from the crack.

Unpacking the Face Terms on the Mesh Skeleton



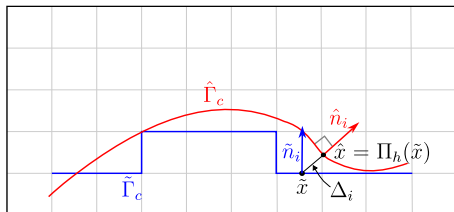
Interior faces split into ordinary faces $\mathcal{E}_{i,o}^h$ and crack faces $\mathcal{E}_c^h$

$$\mathcal{E}_i^h = \mathcal{E}_{i,o}^h \cup \mathcal{E}_c^h$$

$$\mathcal{I}_{\mathcal{E}_i^h}^{\text{flow}}(p, v) = \sum_{e \in \mathcal{E}_i^h} \int_e (\{\{\tilde{q}_i\}\} \tilde{n}_i [v] + [\tilde{q}_i] \tilde{n}_i \{\{v\}\}) d\tilde{s}$$

$$\mathcal{I}_{\mathcal{E}_i^h}^{\text{mech}}(u, p, w) = \sum_{e \in \mathcal{E}_i^h} \int_e (\{\{\tilde{\sigma}_{ij}\}\} \tilde{n}_j [w_i] + [\tilde{\sigma}_{ij}] \tilde{n}_j \{\{w_i\}\}) d\tilde{s}$$

Key Idea: Shift the Laws, Not the Mesh



Gap vector Δ_i from surrogate $\tilde{x}$ to true interface $\hat{x}$

Three steps:

- 1 Replace Γ_c by mesh-fitted surrogate $\tilde{\Gamma}_c$
- 2 Map surrogate points to the true interface by closest-point projection
- 3 Transfer interface data via local Taylor expansion

$$\hat{x} = \Pi_h(\tilde{x}), \quad \Delta_i = \hat{x}_i - \tilde{x}_i$$

All physics is enforced on Γ_c , not on $\tilde{\Gamma}_c$.

Ingredient 1: Decompose the Surrogate Normal

The unresolved crack-face term contains the surrogate normal $\tilde{n}_j$:

$$\mathcal{I}_{\mathcal{E}_c^h}(\Phi, \psi) = \sum_{e \in \mathcal{E}_c^h} \int_e \left(\{\{\tilde{\Phi}_j\}\} \tilde{n}_j \llbracket \psi \rrbracket + \llbracket \tilde{\Phi}_j \rrbracket \tilde{n}_j \{\{\psi\}\} \right) d\tilde{s}$$

$$\tilde{n}_j = (\tilde{n}_k \hat{n}_k) \hat{n}_j + \tilde{n}_j^\perp = \cos \varphi \hat{n}_j + \tilde{n}_j^\perp$$

$$\cos \varphi := \tilde{n}_k \hat{n}_k, \quad \tilde{n}_j^\perp := \tilde{n}_j - (\tilde{n}_k \hat{n}_k) \hat{n}_j, \quad \tilde{n}_k^\perp \hat{n}_k = 0$$

This separates the crack-face contribution into an aligned part along $\hat{n}_j$ and a geometric mismatch part $\tilde{n}_j^\perp$.

Ingredient 2: Transfer True-Crack Data to the Surrogate

A true-crack quantity at $\hat{x}$ is expanded about the surrogate point $\tilde{x}$:

$$\hat{\phi} = \tilde{\phi} + \Delta_j \tilde{\phi}_{,j} + \mathcal{O}(\|\Delta\|^2)$$

For the face field in the crack term, this is used as

$$\tilde{\Phi}_j = \hat{\Phi}_j - \Delta_\ell \tilde{\Phi}_{j,\ell} + \mathcal{O}(\|\Delta\|^2)$$

This rewrites true-interface data as surrogate data plus a first-order Taylor correction, which is what produces the $\mathcal{T}$ term.

Together with the normal split, these are the only ingredients needed for the shifted crack-face formulation.

Shift the Crack-Face Term

Combining normal decomposition and Taylor transfer gives:

Before shifting

$$\mathcal{I}_{\mathcal{E}_c^h}(\Phi, \psi) = \sum_{e \in \mathcal{E}_c^h} \int_e (\{\{\tilde{\Phi}_j\}\} \tilde{n}_j \llbracket \psi \rrbracket + \llbracket \tilde{\Phi}_j \rrbracket \tilde{n}_j \{\{\psi\}\}) d\tilde{s}$$

Shift by decomposing $\tilde{n}_j = \cos \varphi \hat{n}_j + \tilde{n}_j^\perp$ and Taylor-transferring true-crack data back to $\tilde{\Gamma}_c$.

After shifting

$$\{\{\tilde{\Phi}_j\}\} \tilde{n}_j = \underbrace{\cos \varphi \{\{\hat{\Phi}_j\}\} \hat{n}_j}_{\mathcal{C}} + \underbrace{\{\{\tilde{\Phi}_j\}\} \tilde{n}_j^\perp}_{\mathcal{N}} - \underbrace{\cos \varphi \{\{\Delta_\ell \tilde{\Phi}_{j,\ell}\}\} \hat{n}_j}_{\mathcal{T}}$$

$$\mathcal{I}_{\tilde{\Gamma}_c} = \mathcal{C}_{\tilde{\Gamma}_c} + \mathcal{N}_{\tilde{\Gamma}_c} + \mathcal{T}_{\tilde{\Gamma}_c}$$

Same template for flow and mechanics; only the constitutive closure changes.

Specific Forms with Interface Unknowns

Flow. With $\hat{q}^{n,\pm} := \hat{q}_j^\pm \hat{n}_j^\pm$,

$$\begin{aligned} \mathcal{I}_{\mathcal{E}_c^h}^{\text{flow}}(p, v) = & \sum_{e \in \mathcal{E}_c^h} \int_e \left\{ \frac{1}{2} \cos \varphi \llbracket \hat{q}^n \rrbracket \llbracket v \rrbracket + 2 \cos \varphi \{ \{ \hat{q}^n \} \} \{ \{ v \} \} \right. \\ & + \{ \{ \tilde{q}_j \} \} \tilde{n}_j^\perp \llbracket v \rrbracket + \llbracket \tilde{q}_j \rrbracket \tilde{n}_j^\perp \{ \{ v \} \} \\ & \left. - \cos \varphi \{ \{ \Delta_\ell \tilde{q}_{j,\ell} \} \} \hat{n}_j \llbracket v \rrbracket - \cos \varphi \llbracket \Delta_\ell \tilde{q}_{j,\ell} \rrbracket \hat{n}_j \{ \{ v \} \} \right\} d\tilde{s}, \\ \llbracket \hat{q}^n \rrbracket = & -2T_n(\hat{x}) \llbracket \tilde{p} + \Delta_\ell \tilde{p}_{,\ell} \rrbracket, \quad \{ \{ \hat{q}^n \} \} = 0. \end{aligned}$$

Mechanics. With $\hat{t}_i^\pm = \hat{\sigma}_{ij}^\pm \hat{n}_j^\pm$,

$$\begin{aligned} \mathcal{I}_{\mathcal{E}_c^h}^{\text{mech}}(u, p, w) = & \sum_{e \in \mathcal{E}_c^h} \int_e \left\{ \frac{1}{2} \cos \varphi \llbracket \hat{t}_i \rrbracket \llbracket w_i \rrbracket + 2 \cos \varphi \{ \{ \hat{t}_i \} \} \{ \{ w_i \} \} \right. \\ & + \{ \{ \tilde{\sigma}_{ij} \} \} \tilde{n}_j^\perp \llbracket w_i \rrbracket + \llbracket \tilde{\sigma}_{ij} \rrbracket \tilde{n}_j^\perp \{ \{ w_i \} \} \\ & \left. - \cos \varphi \{ \{ \Delta_\ell \tilde{\sigma}_{ij,\ell} \} \} \hat{n}_j \llbracket w_i \rrbracket - \cos \varphi \llbracket \Delta_\ell \tilde{\sigma}_{ij,\ell} \rrbracket \hat{n}_j \{ \{ w_i \} \} \right\} d\tilde{s}, \\ \{ \{ \hat{t}_i \} \} = & 0, \quad \llbracket \hat{t}_i \rrbracket = 2 \left(K_{ij}(\hat{x}) \llbracket \tilde{u}_j + \Delta_\ell \tilde{u}_{j,\ell} \rrbracket + H_{ij}(\hat{x}) \llbracket \dot{\tilde{u}}_j + \Delta_\ell \dot{\tilde{u}}_{j,\ell} \rrbracket \right). \end{aligned}$$

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Two Strategies for Interface Enforcement

Weak Enforcement

- Constitutive law substituted **directly into crack-face integrals**
- **Integral enforcement** during assembly
- No additional **interface unknowns**

Strong Enforcement

- Fluxes/tractions kept as **additional interface unknowns** on $\tilde{\Gamma}_c$
- Constitutive law enforced **algebraically / pointwise** at interface nodes
- Convenient for nonlinear interface laws

Same bulk operators and shifted geometric corrections;
only the constitutive closure is treated differently.

Semidiscrete System: Weak Enforcement

$$\underbrace{\begin{bmatrix} \mathbf{M}_p & \mathbf{C} \\ \mathbf{0} & \mathbf{D}_u^c \end{bmatrix}}_{\mathbf{E}} \begin{bmatrix} \dot{\mathbf{p}} \\ \dot{\mathbf{u}} \end{bmatrix} + \underbrace{\begin{bmatrix} \mathbf{K}_p + \mathbf{K}_p^c + \mathbf{S}_p^c + \mathbf{R}_p^c & \mathbf{0} \\ -\mathbf{C}^T + \mathbf{S}_{up}^c + \mathbf{R}_{up}^c & \mathbf{K}_u + \mathbf{K}_u^c + \mathbf{S}_{uu}^c + \mathbf{R}_{uu}^c \end{bmatrix}}_{\mathbf{K}} \begin{bmatrix} \mathbf{p} \\ \mathbf{u} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_p \\ \mathbf{f}_u \end{bmatrix}$$

- $(\mathbf{K}^c, \mathbf{D}^c)$: constitutive interface blocks ($\mathcal{C}$)
- $\mathbf{S}^c$: Taylor corrections ($\mathcal{T}$) $\mathbf{R}^c$: normal-mismatch corrections ($\mathcal{N}$)
- Index-1 DAE solved with adaptive SDIRK2

Semidiscrete System: Strong Enforcement

$$\mathbf{E}_{\text{bulk}} \dot{\mathbf{z}} + \mathbf{K}_{\text{bulk}} \mathbf{z} + \mathbf{G} \boldsymbol{\lambda} = \mathbf{f}$$

with algebraic closure at each interface node:

$$\boldsymbol{\lambda} = \mathcal{C}(\mathbf{p}, \mathbf{u}, \dot{\mathbf{u}})$$

- $\boldsymbol{\lambda} = [\hat{\mathbf{q}}^{n,\pm}, \hat{\mathbf{t}}^{\pm}]^T$: interface unknowns
- $\mathbf{G}$: geometry-dependent coupling blocks, assembled once
- Constitutive law $\mathcal{C}$ updated pointwise each time step

Postprocessing: Results on the true crack Γ_c

Solve on $\tilde{\Gamma}_c$ — report on Γ_c

- 1 Reconstruct smooth gradients** (for fluxes & stresses)

$$Mg_j = G_j f$$

Enables $\tilde{q}_i, \tilde{\sigma}_{ij}$ on $\tilde{\Gamma}_c$

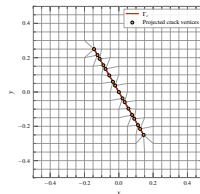
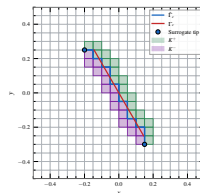
- 2 Taylor-transfer p, u to true crack** (key step)

$$f^h(\hat{x}) \approx f^h(\tilde{x}) + \Delta_j f^h_{,j}(\tilde{x})$$

Maps pressure and displacement from $\tilde{\Gamma}_c$ to Γ_c

- 3 Recompute fluxes and tractions** on Γ_c

$\hat{q}_n = \tilde{q}_i \hat{n}_i, \quad \hat{t}_j = \tilde{\sigma}_{ij} \hat{n}_i$
All reported results are evaluated on Γ_c , not on $\tilde{\Gamma}_c$.



Common Numerical Setup

Discretization

- 2D plane strain
- Structured quadrilateral meshes
- MINI-type $\mathbb{Q}_1$ -bubble / $\mathbb{Q}_1$
- Mesh sequence
 $n \in \{20, 40, 80, 160, 320\}$

Comparison Strategy

- Weak and strong enforcement compared on single-crack cases

One common framework, then increasingly challenging geometries.

Boundary Conditions

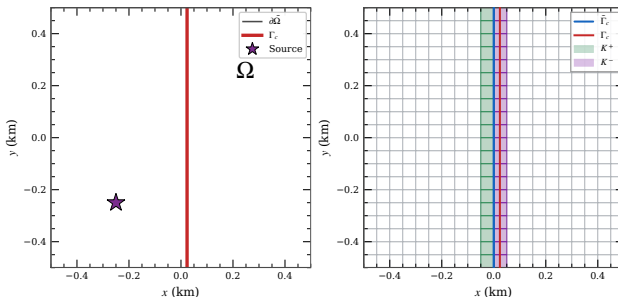
- $u_x = u_y = 0$ on left/right
- Traction-free top/bottom
- $p = 0$ on all outer boundaries

Default Single-Crack Interface (Tests 1–2)

- large K_{ij} : effectively glued
- $T_n = 0$: impermeable

Test 1: Offset Mesh-Aligned Crack

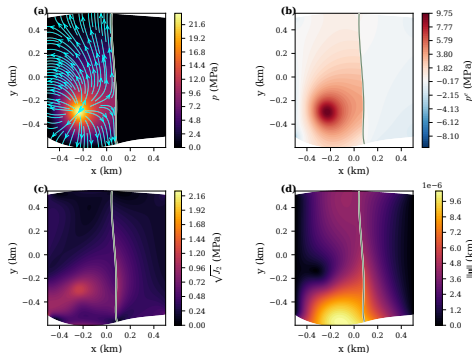
Test 1: Offset Crack — Setup



- Crack offset by half an element width
- Surrogate $\tilde{\Gamma}_c$ aligned with element faces; gap Δ_i purely horizontal

Test 1: Offset Mesh-Aligned Crack

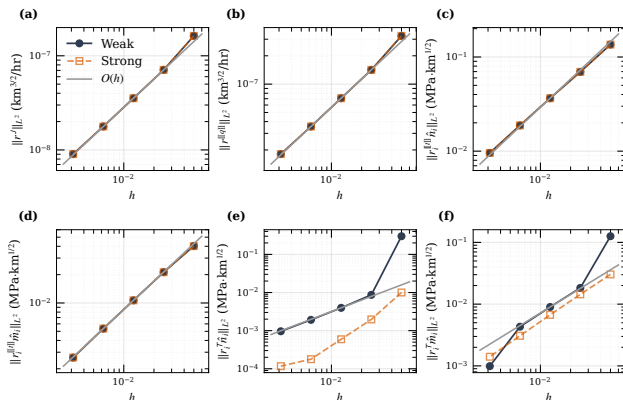
Test 1: Solution Fields



Both enforcement strategies yield identical solution fields at this resolution.

Test 1: Offset Mesh-Aligned Crack

Test 1: Interface Residual Convergence



Curve meanings

- $r[q]$: flux balance
- r^f : flow constitutive law
- $r[t]$: traction balance
- r^T : spring law

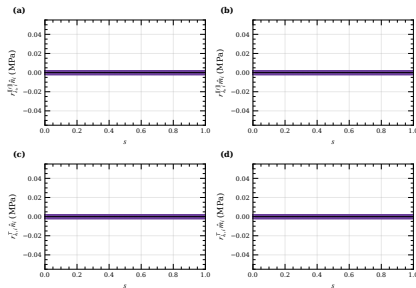
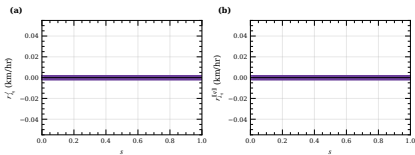
All curves are post-processed residuals evaluated on Γ_C .

Interface conditions converge optimally on Γ_C .

Test 1: Offset Mesh-Aligned Crack

What Strong Enforcement Satisfies Exactly

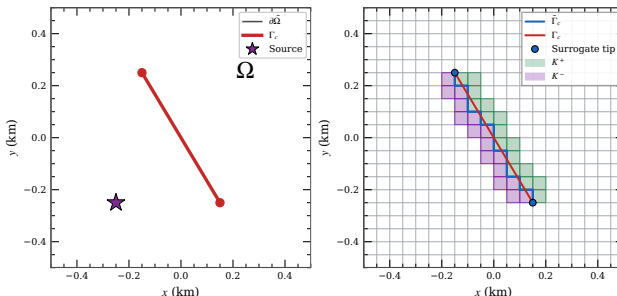
Algebraic closure residuals at interface DOFs are zero.



- λ_q, λ_t satisfy the constitutive laws pointwise at interface nodes

Test 2: Embedded Angled Crack (30 degrees)

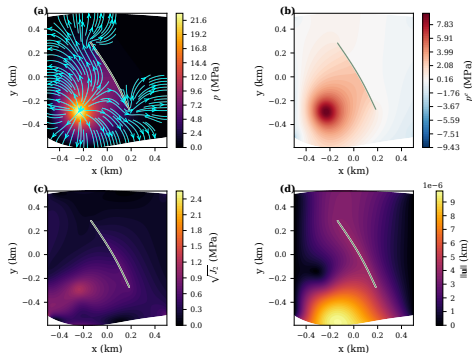
Test 2: Embedded Angled Crack — Setup



- Crack rotated 30° from vertical; both tips in the domain interior
- Surrogate is a **staircase** approximation; full SIM machinery exercised

Test 2: Embedded Angled Crack (30 degrees)

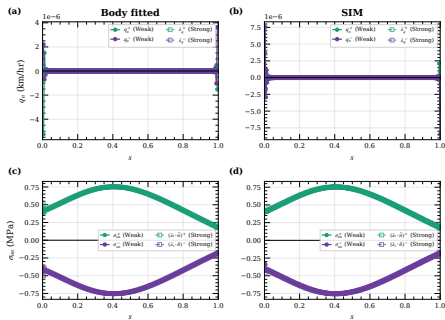
Test 2: Solution Fields



Embedded crack confines the pressure discontinuity to the central domain region.

Test 2: Embedded Angled Crack (30 degrees)

Test 2: Body-Fitted vs SIM — Flux & Normal Traction

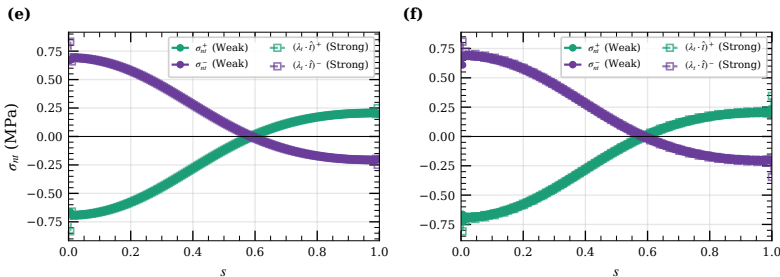


(a,b) Normal flux q_n ; (c,d) Normal traction σ_{nn} .

SIM matches the body-fitted solution in the crack interior.

Test 2: Embedded Angled Crack (30 degrees)

Test 2: Body-Fitted vs SIM — Tangential Traction

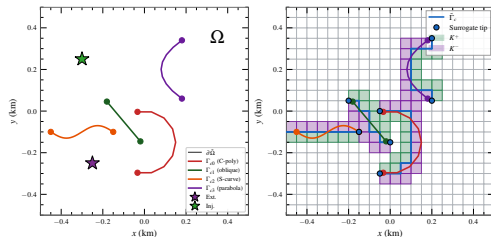


(e,f) Tangential traction σ_{nt} .

Largest discrepancy is localised at crack tips and diminishes under refinement.

Test 3: Multi-Crack Configuration

Test 3: Multi-Crack Configuration



4 cracks: C-shaped, oblique, S-curve, parabolic arc.
Injection-extraction doublet drives the test.

All cracks handled on a single structured mesh.

No special algorithmic treatment beyond supplying crack descriptors.

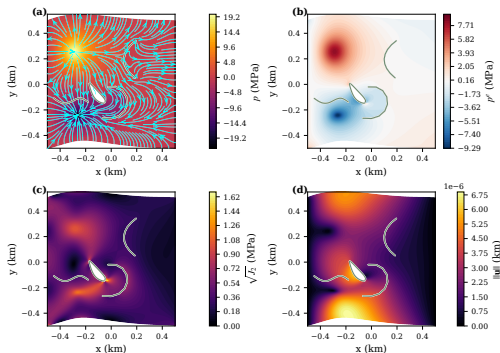
Interface heterogeneity

- Γ_{c0}, Γ_{c3} : impermeable, welded
- Γ_{c1} : compliant ($k_n = 10^4$)
- Γ_{c2} : semi-permeable ($T_n = 5$)

4 non-intersecting cracks on a single structured mesh.

Test 3: Multi-Crack Configuration

Test 3: Multi-Crack Solution Fields



Flow is redirected by impermeable barriers; the compliant crack opens mechanically under the pressure differential.

Summary

- **Unified SIM for coupled poroelastic interfaces**
- **Strong enforcement satisfies interface laws pointwise**
- **Accurate in the crack interior** — remaining visible errors are concentrated near staircase-treated tips
- **Multiple non-intersecting cracks** on one structured mesh with heterogeneous properties

No cut cells, no enrichments, minimal intrusion into standard FE infrastructure.

But What About Realistic Contact?

What SIM solved today

- Geometry need not fit the crack
- Interface laws are transferred to Γ_c
- Standard FE data structures remain usable

What remains

- Crack faces may open or close
- Contact reactions are unilateral
- Friction switches between stick and slip

Next lecture

From smooth interface closures to nonsmooth contact laws:

$$g_n \geq 0, \quad r_n \geq 0, \quad g_n r_n = 0, \quad \|r_t\| \leq \mu r_n.$$

The next difficulty is not only geometry. It is **changing regime**.

Thank you!

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